Jianhao Zhang, Ph.D.

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Research Interests

My research characterizes and quantifies how **clouds** respond to changes in **aerosol** loading and meteorological conditions driven by human activity and natural **climate** variability. Building on this, I seek to understand how these cloud responses shape the increasing Earth's energy imbalance and constrain the viability and risks of proposed climate interventions. To address these multiscale questions, I leverage satellite and field observations, atmospheric and Earth system modeling, and data-driven approaches to extract physical understanding of the processes linking clouds to a changing climate.

Professional Appointment

May 2026 – present — **Research Scientist III,** NOAA Chemical Sciences Laboratory & CIRES at the University of Colorado Boulder *(Clouds, Aerosol, & Climate Group)*

May 2023 – Apr 2026 — **Research Scientist II,** NOAA Chemical Sciences Laboratory & CIRES at the University of Colorado Boulder *(Clouds, Aerosol, & Climate Group)*

Sept. 2021 – Apr. 2023 — **Research Scientist I,** NOAA Chemical Sciences Laboratory & CIRES at the University of Colorado Boulder *(Clouds, Aerosol, & Climate Group)*

Sept. 2020 – Aug. 2021 — **NRC Postdoc fellowship** hosted by Graham Feingold at NOAA Chemical Sciences Laboratory

Education

2020 — **Ph.D., University of Miami, Miami, FL, USA** in Meteorology & Physical Oceanography.
Dissertation title: *The interactions between light-absorbing smoke and marine boundary layer clouds over the remote southeast Atlantic.*
Committee: *Paquita Zuidema (Chair), Brian Mapes, Brian Soden, Cassandra Gaston, David Turner & Takanobu Yamaguchi*

2014 — **B.S., Florida State University, Tallahassee, FL, USA** in Meteorology.
Magna cum laude; Minor in Mathematics.

Funded Grants

2026-ongoing — **Co-I.:** Assessing the detectability of small-scale stratospheric aerosol injections (**DOD** ESTCP Natural Hazards Project NH26-9821)

2025-2026 — **P. I.:** A Framework for Assessing SRM Detectability for Informed Decision-Making (**Reflective**, $162,768)

2023-2026 — **Co-I.:** Aerosol-Cloud Interactions Centered on MAGIC: Insights from Measurements and Lagrangian Large Eddy Simulation (**DOE** Atmospheric System Research, $670,445)

2023-2024 — **P.I. for CIRES:** Exploring Aerosol-Cloud Interactions in Geophysical Variable Spaces using NASA-ACTIVATE Observations (**NASA** Earth Venture Suborbital-3, $350,000)

2020-2023 — **Co-I.:** Evaluating Biases in Aerosol-Cloud Interaction Metrics using ARM Data and Models (**DOE** Atmospheric System Research, $510,472)

2020-2021 — **P. I.:** National Research Council Fellowship Award (**National Academies of Sciences, Engineering, Medicine**, $62,000)

Awards, Fellowships, & Honors

- **NOAA Boulder Outreach Gold Star Award, 2025**
- **Reflective Fellowship, 2025**, Reflective Inc.
- **National Research Council Research Associateship Fellowship, 2020**, National Academies of Sciences, Engineering, and Medicine
- **NASA's Group Achievement Awards** (ACTIVATE Team, 2023; ORACLES Team, 2019)
- **Meteorology & Physical Oceanography Program Best Ph.D. Dissertation, 2020**
- **University of Miami Graduate Fellowship, 2014** (highest award of the graduate school)

Teaching, Mentoring, & Outreach

Teaching
The Program for Exploratory Studies as *Guest lecturer*, CU Boulder (Apr 2026)
Aerosol-Cloud-Climate Interactions as *Guest lecturer*, TU Delft (June 2024)
Cloud Physics and Radiative Transfer as *Teaching Assistant*, University of Miami (2016)
Introduction to Weather and Climate as *Teaching Assistant*, University of Miami (2015)
AP calculus and undergrad statistics as *Math Tutor*, Miami (2015-2020)

Mentoring
Tyler Tatro, Ph.D. student at University of Miami (2022-2026)
Alexander J. Thompson, Research Scientist at NOAA/PSL (2024-2025)
Danyan Leng, Ph.D. student at University of Colorado Boulder (2025-2026)
Daisy Kerr, NOAA Hollings Scholar, Physics undergrad at UC Berkeley (Summer 2026)

Outreach
CIRES Speaker, CIRES Science Pathways Program (2025-)
Scientist Panel, Environmental studies class at Boulder High School (2025)
Panelist, CIRES Graduate Student Workshop (2025)
Invited Scientist, Boulder Valley Rotary Club Weekly Discussion (2025)
Certificate, Dialogic Skills Workshop, University of Colorado Boulder (2025)
Science vetting, Children's book *UP, UP HIGH*
Evaluator, CU Boulder RECCS Symposium 2024; GLOBE International Virtual Science Symposium 2022; Climate Literacy and Energy Awareness Network 2022

Leadership & Service

Reviewer
Department of Energy (DOE) Atmospheric System Research (ASR)
University of Colorado AB Nexus Program
NOAA Hollings Undergraduate Scholarship
Peer-reviewer for 18 major academic journals

Committee
AGU Dewan Scholarship (2026-present)
CIRES Members' Council (2025-present)
CIRES Mentoring Program (2024-present, Chair for 2025-2026)
NOAA OAR subject matter expert in *Satellite Data* (2022-present)
University of Miami RSMAS Student Seminar Committee (2016)

Convener
AMS's, *Advances in Assessing Aerosol Impacts on Clouds Through AI and Advanced Statistical Techniques*, Primary Convener since 2026
AMS's 17th Conference on Atmospheric Radiation, 2026, Madison, WI
AGU 2024, *Advances in Cloud and Precipitation Processes: Integrating Observations, Modeling, and Theory*, Washington D.C.
AMS's 16th Conference on Cloud Physics, 2022, Madison, WI

Discussion Lead
2025 Gordon Research Seminar on Radiation and Climate, Lewiston, ME
University of Washington Inaugural MCB Workshop, 2025, Leavenworth, WA

Field campaign
NASA ORACLES-2016, Swakopmund, Namibia, Sept. 2016
NASA ORACLES-2017, São Tomé, São Tomé and Príncipe, Aug. 2017

Invited Presentations

May 2026 — **Libera Science Team Meeting, Berkeley, CA,** *Earth's East-West Albedo Symmetry.*

Mar. 2026 — **NOAA Chemical Sciences Laboratory Seminar, Boulder, CO,** *A cloudy story of Earth's radiation budget.*

May 2025 — **AGU AS Early-Career Seminar, Online,** *Global climate and air quality implications of regional emission shift.*

Apr. 2025 — **Boulder Valley Rotary Club Weekly Meeting, Boulder, CO,** *How to measure the shade of clouds that are no longer there.*

- **Climate Dynamics & Impacts, Vecchi/Soden Joint Group Meeting, Princeton University and University of Miami, Online,** *Large radiative forcing from the 2020 shipping fuel regulation is hard to detect: Implications for Marine Cloud Brightening.*
- **University of Washington Inaugural MCB Program Workshop, Leavenworth, WA, Session co-lead,** *Identifying conditions amenable to cloud brightening/MCB.*
- **NOAA Science Seminar Series, Online,** *AI Applications in Earth System and Climate Science: Aerosols and Air Quality.*

Nov. 2024 — **NOAA Advancing Innovative Research Seminar Series, Online,** *Large radiative forcing from the 2020 shipping fuel regulation is hard to detect.*

Jun. 2024 — **TU Delft, Delft, Netherlands,** *On the viability of Marine Cloud Brightening: Albedo susceptibility, cloud adjustment, and detectability.*

Oct. 2023 — **Brookhaven National Laboratory, Long Island, NY,** *Aerosol-cloud interactions in marine warm clouds and implications for Marine Cloud Brightening.*

Dec. 2022 — **2022 AGU Fall Meeting, Chicago, IL,** *On the Conditionality of Marine Low Cloud Albedo Susceptibility: from Meteorological Conditions to Spatiotemporal Scales.*

Media & Press

- **Phys.org**, *Understanding Earth's hidden east-west symmetry could improve climate models*
- **New Scientist**, *Earth has a mysterious triple symmetry that may influence its climate*
- **CIRES News**, *Clouds shape a newly identified east–west symmetry in Earth's reflected sunlight*
- **Reflective blog post**, *Ensuring continuity for atmospheric research*
- **CIRES Spotlights**, *Flying high in the sky*
- **NOAA CPO News**, *The unintended consequences of reducing sulfur emissions from ships*
- **NOAA Research**, *Scientists turn to artificial intelligence to assess the warming effect of reduced pollution*
- **CIRES News**, *Scientists turn to artificial intelligence to assess the warming effect of reduced pollution*
- **SciTechDaily**, *Cleaner Ships, Hotter Earth: The Unexpected Climate Twist*
- **Mongabay**, *Shipping emissions reduction sheds light on marine cloud geoengineering*
- **DOE ARM NEWS**, *Smoke Studies: Crucial Cloud-Deck Science*

References

Graham Feingold
Senior Science Advisor, Renaissance Philanthropy
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Robert Wood
Professor, University of Washington
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Paquita Zuidema
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Johannes Mülmenstädt
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Publications

Peer-reviewed

23 M. S. Diamond, A. S. Ackerman, A. Baró Pérez, et al., "Drizzle drives model differences in the Southeastern Atlantic Stratocumulus Transitions with Aerosol-Rain-Radiation interactions (SEA STARR) large eddy simulation intercomparison project," *Atmos. Chem. Phys.*, 2026, *under review*. DOI: 10.5194/egusphere-2026-5350

22 T. Dror, **J. Zhang**, and G. Feingold, "Amazon darkening mitigated by all-sky biophysical feedbacks," *Nature Communications*, 2026, *under review*.

21 H. He, B. J. Soden, G. A. Vecchi, R. J. Kramer, **J. Zhang**, and C. Park, "State dependence of aerosol forcing and its implications for climate response," *Atmos. Chem. Phys.*, 2026, *under review*. DOI: 10.5194/egusphere-2026-6648

20 H. He, B. J. Soden, G. A. Vecchi, R. J. Kramer, **J. Zhang**, and C. Park, "Technical note: Aerosol forcing decomposition methods and recommendations," *Atmos. Chem. Phys.*, 2026, *under review*. DOI: 10.5194/egusphere-2026-6647

19 **J. Zhang**, J. Gristey, and G. Feingold, "Earth's East-West albedo symmetry," *Nature*, vol. 654, pp. 676–682, 2026. DOI: 10.1038/s41586-026-10624-2

18 **J. Zhang**, D. Painemal, T. Dror, J.-S. Lim, A. Sorooshian, and G. Feingold, "Inferring processes governing cloud transition during mid-latitude marine cold-air outbreaks from satellite," *Atmos. Chem. Phys.*, vol. 26, no. 9, pp. 6015–6034, 2026. DOI: 10.5194/acp-26-6015-2026

17 G. Feingold, F. Glassmeier, **J. Zhang**, and F. Hoffmann, "Opinion: Inferring process from snapshots of cloud systems," *Atmos. Chem. Phys.*, vol. 25, no. 18, pp. 10 869–10 885, 2025. DOI: 10.5194/acp-25-10869-2025

16 **J. Zhang**, Y.-S. Chen, E. Gryspeerdt, T. Yamaguchi, and G. Feingold, "Radiative forcing from the 2020 shipping fuel regulation is large but hard to detect," *Commun. Earth Environ.*, vol. 6, no. 18, pp. 1–11, 2025. DOI: 10.1038/s43247-024-01911-9

15 Y.-S. Chen, **J. Zhang**, F. Hoffmann, T. Yamaguchi, F. Glassmeier, X. Zhou, and G. Feingold, "Diurnal evolution of non-precipitating marine stratocumuli in a large-eddy simulation ensemble," *Atmos. Chem. Phys.*, vol. 24, no. 22, pp. 12 661–12 685, 2024. DOI: 10.5194/acp-24-12661-2024

14 G. Feingold, V. Ghate, L. M. Russell, et al., "Physical science research needed to evaluate the viability and risks of marine cloud brightening," *Science Advances*, vol. 10, no. 12, eadi8594, 2024. DOI: 10.1126/sciadv.adi8594

13 **J. Zhang**, Y.-S. Chen, T. Yamaguchi, and G. Feingold, "Cloud water adjustments to aerosol perturbations are buffered by solar heating in non-precipitating marine stratocumuli," *Atmos. Chem. Phys.*, vol. 24, no. 18, pp. 10 425–10 440, 2024. DOI: 10.5194/acp-24-10425-2024

12 C. Howes, P. E. Saide, H. Coe, et al., "Biomass-burning smoke's properties and its interactions with marine stratocumulus clouds in WRF-CAM5 and southeastern Atlantic field campaigns," *Atmos. Chem. Phys.*, vol. 23, no. 21, pp. 13 911–13 940, 2023. DOI: 10.5194/acp-23-13911-2023

11 **J. Zhang** and G. Feingold, "Distinct regional meteorological influences on low-cloud albedo susceptibility over global marine stratocumulus regions," *Atmos. Chem. Phys.*, vol. 23, no. 2, pp. 1073–1090, 2023. DOI: 10.5194/acp-23-1073-2023

10 P. A. Barrett, S. J. Abel, H. Coe, et al., "Intercomparison of airborne and surface-based measurements during the clarify, oracles and lasic field experiments," *Atmos. Meas. Tech.*, vol. 15, no. 21, pp. 6329–6371, 2022. DOI: 10.5194/amt-15-6329-2022

9. M. S. Diamond, P. E. Saide, P. Zuidema, et al., "Cloud adjustments from large-scale smoke–circulation interactions strongly modulate the southeastern atlantic stratocumulus-to-cumulus transition," *Atmos. Chem. Phys.*, vol. 22, no. 18, pp. 12 113–12 151, 2022, (ACP highlight). DOI: 10.5194/acp-22-12113-2022

8. **J. Zhang**, X. Zhou, T. Goren, and G. Feingold, "Albedo susceptibility of northeastern pacific stratocumulus: The role of covarying meteorological conditions," *Atmos. Chem. Phys.*, vol. 22, no. 2, pp. 861–880, 2022. DOI: 10.5194/acp-22-861-2022

7. **J. Zhang** and P. Zuidema, "Sunlight-absorbing aerosol amplifies the seasonal cycle in low-cloud fraction over the southeast atlantic," *Atmos. Chem. Phys.*, vol. 21, no. 14, pp. 11 179–11 199, 2021. DOI: 10.5194/acp-21-11179-2021

6. X. Zhou, **J. Zhang**, and G. Feingold, "On the importance of sea surface temperature for aerosol-induced brightening of marine clouds and implications for cloud feedback in a future warmer climate," *Geophys. Res. Lett.*, vol. 48, no. 24, e2021GL095896, 2021. DOI: https://doi.org/10.1029/2021GL095896

5. S. J. Abel, P. A. Barrett, P. Zuidema, **J. Zhang**, M. Christensen, F. Peers, J. W. Taylor, I. Crawford, K. N. Bower, and M. Flynn, "Open cells exhibit weaker entrainment of free-tropospheric biomass burning aerosol into the south-east Atlantic boundary layer," *Atmos. Chem. Phys.*, vol. 20, no. 7, pp. 4059–4084, 2020. DOI: 10.5194/acp-20-4059-2020

4. **J. Zhang** and P. Zuidema, "The diurnal cycle of the smoky marine boundary layer observed during August in the remote southeast Atlantic," *Atmos. Chem. Phys.*, vol. 19, no. 23, pp. 14 493–14 516, 2019, (ACP highlight). DOI: 10.5194/acp-19-14493-2019

3. A. S. Chandra, P. Zuidema, S. Krueger, A. Kochanski, S. P. de Szoeke, and **J. Zhang**, "Moisture distributions in tropical cold pools from equatorial Indian ocean observations and cloud-resolving simulations," *J. Geophys. Res. Atmos.*, vol. 123, no. 20, pp. 11, 445–11, 465, 2018. DOI: 10.1029/2018JD028634

2. **J. Zhang**, P. Zuidema, D. D. Turner, and M. P. Cadeddu, "Surface-based microwave humidity retrievals over the equatorial Indian ocean: Applications and challenges," *J. Appl. Meteor. Climatol.*, vol. 57, no. 8, pp. 1765–1782, 2018. DOI: 10.1175/JAMC-D-17-0301.1

1. P. Zuidema, A. J. Sedlacek III, C. Flynn, S. Springston, R. Delgadillo, **J. Zhang**, A. C. Aiken, A. Koontz, and P. Muradyan, "The Ascension island boundary layer in the remote southeast Atlantic is often smoky," *Geophys. Res. Lett.*, vol. 45, no. 9, pp. 4456–4465, 2018. DOI: 10.1002/2017GL076926

Other Publications

1. **J. Zhang** and G. Feingold, "Physical Science of Marine Cloud Brightening: Knowledge and Gaps," in *Topical Group on the Physics of Climate*, (invited article), American Physical Society, October 2024, pp. 1–4. URL: https://engage.aps.org/gpc/resources/newsletters